

Photosynthesis

Photosynthesis is the process by which green plants, algae, and some bacteria make their own food using sunlight, carbon dioxide, and water. In this process, plants use sunlight energy captured by a green pigment called chlorophyll present in their leaves. They take in carbon dioxide from the air through small openings called stomata and absorb water from the soil through their roots. Using sunlight energy, plants convert carbon dioxide and water into glucose (a type of sugar), which provides energy for growth and development. Oxygen is produced as a by-product and released into the atmosphere. The overall chemical equation of photosynthesis is: Carbon dioxide + Water → Glucose + Oxygen. Photosynthesis is very important because it provides food for all living organisms either directly or indirectly. It also releases oxygen, which is essential for breathing and survival of most living beings.